

WSI Viewer Release Cheat Sheet

Development -> Staging -> Production rehearsal -> Production -> Tag

Environments

Environment	Port	Runtime / identity	Data boundary
Development	8081	Live source; red banner	Deidentified development images and annotations
Staging	8082	Candidate JAR; yellow banner	Deidentified staging images and annotations
Rehearsal	8083	Exact staging JAR; production mode; loopback only	Deidentified production-marked images; rehearsal annotations
Production	8080	Frozen validated JAR; no banner	Authorized clinical images and production annotations

Normal release

1. Development gate

```
cd /Users/dm026/Downloads/wsi-server_works
git status --short
./mvnw clean test
node --test src/test/js/*.test.js
./ops/tests/run.sh
git push origin feature/multichannel-viewer
```

STOP: Git must be clean and every required test must pass.

2. Build and install staging

```
./ops/wsi-release stage --dry-run
./ops/wsi-release stage --step
./ops/wsi-release verify staging
wsi staging status
```

Browser 8082: yellow banner; deidentified slides; viewer, channels, annotations and exports; no 403, 500 or unhandled JavaScript errors.

3. Rehearse production mode

```
./ops/wsi-release rehearse --dry-run
./ops/wsi-release rehearse --step
./ops/wsi-release verify rehearsal
wsi rehearsal status
```

Browser 8083: no banner; production layout; exact staging artifact; deidentified rehearsal data only. Port 8080 remains available.

4. Promote exact rehearsed candidate

```
./ops/wsi-release verify staging
./ops/wsi-release verify rehearsal
./ops/wsi-release promote --dry-run
./ops/wsi-release promote --step
./ops/wsi-release verify production
```

Type PROMOTE only when staging and rehearsal commit/SHA-256 match and the rollback backup verifies.

5. Tag after production validation

```
./ops/wsi-release tag production-YYYY-MM-DD-description
```

Browser 8080: normal production layout/title; authorized slides; viewer, channels, annotations and exports; no security, layout or persistence regression; other environments remain isolated.

Troubleshooting and Recovery

Status and verification

```
wsi production status
wsi staging status
wsi rehearsal status
wsi development status
./ops/wsi-release status
./ops/wsi-release verify staging
./ops/wsi-release verify rehearsal
./ops/wsi-release verify production
```

Logs

```
wsi production logs
wsi staging logs
wsi rehearsal logs
wsi development logs
```

Press Control-C to stop following a log.

Rollback

```
./ops/wsi-release history
./ops/wsi-release rollback --step
./ops/wsi-release verify production
```

Rollback restores the prior JAR, metadata, checksum and configuration. It never restores or overwrites annotations.

Troubleshooting modes

```
./ops/wsi-release stage --dry-run
./ops/wsi-release stage --step --verbose
./ops/wsi-release rehearse --dry-run
./ops/wsi-release promote --dry-run
./ops/wsi-release history
```

--dry-run: plan only. --step: Enter runs, p repeats, q exits safely. Use ./ops/wsi-release if the installed command is missing. Never bypass PROMOTE or ROLLBACK.

Stop conditions

- Dirty Git tree or failed required test
- Git HEAD differs from staging commit
- Recorded/calculated SHA-256 differ
- Staging and rehearsal differ
- Environment identity, marker, root or port mismatch
- Production backup validation fails
- New 403, 500, unhandled JavaScript error, lost annotation, blank image or routine delay

Do not repair a mismatch by editing build metadata or checksum files.

Marker rule

Identity	Required marker
development	.wsi-environment-development
staging	.wsi-environment-staging
production / rehearsal	.wsi-environment-production

Timing: Image-switch total ends at OpenSeadragon open and may not include all visible tiles. Use browser Network timing and server logs for repeatable delays.

Privacy: Never copy production slides into development, staging or rehearsal. Verify filenames, metadata, embedded labels, thumbnails, macro images and associated files. Never synthesize a missing embedded label/thumbnail from diagnostic pixels.

Exactly one environment marker must exist. A missing, multiple or cross-environment marker prevents startup before the server accepts requests.